

WEEK-9

⇒ Multivariable Function

Directional Derivative

Let $f(x_1, x_2, \dots, x_n)$ be a function defined on a domain $D \subset \mathbb{R}^n$.

The directional derivative of f in the direction of the unit vector u is the function denoted by $f_u(x)$ and defined as

$$f_u(x) = \lim_{h \rightarrow 0} \frac{f(x + hu) - f(x)}{h}$$

$x \rightarrow$ Given point

$u \rightarrow$ Given unit vector

Where its differentiable we can use $D_u f = \nabla f \cdot u$
Point exist

Graded

Q. Find out the directional derivative of the function $f(x, y) = x^2 y^3$ at $(1, 4)$, in the direction of the vector $(0, 3)$.

Sol.

$$f(x, y) = x^2 y^3$$

Gradient : $\nabla f = (2xy^3, 3x^2 y^2)$
along x and y
At $(1, 4) = (128, 48)$

Normalize : $(0, 3) \rightarrow$ Unit vector $\rightarrow (0, 1) \rightarrow u$
direction

Directional derivative $\Rightarrow D_u f = \nabla f \cdot u = (128, 48) \cdot (0, 1) = \boxed{48}$

Graded

If the rate of change of the direction of the vector (l, m) is $\frac{1}{\sqrt{l^2 + m^2}} [ka + lb]$, Price of the Product along the material is a and the Price of raw is b , the find the value of transformation of the product to the market

Sol.

① $f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$

$f_x = 2x + y, f_y = x + 2y$

So: $\nabla f = (2x + y, x + 2y) \rightarrow$ at a & b $\nabla f(a, b) = (2a + b, a + 2b)$

② Unit vector : $u = \frac{(l, m)}{\sqrt{l^2 + m^2}}$

$D_u f = \nabla f \cdot u$
 $\Rightarrow \frac{1}{\sqrt{l^2 + m^2}} [(2a + b) \cdot l + (a + 2b) \cdot m]$

③ Group terms :

$$\Rightarrow \frac{1}{\sqrt{1+m^2}} \left[\underbrace{(2+m)}_k a + \underbrace{(1+2m)}_l b \right]$$

So,

$$k = 2+m$$

$$l = 1+2m$$

$$\text{Find } 2k-l = 2(2+m) - (1+2m) = \textcircled{3}$$

ungraded

★ Q. Consider the $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as : $f(x,y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & x, y \neq 0 \\ 0 & x=y=0 \end{cases}$

Find the directional derivative of f at $(0,0)$ in the direction of the vector $u = \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right)$.

Sol.

$$u = \frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j}$$

$$\text{Point} = (0,0)$$

$$\lim_{h \rightarrow 0} \frac{f\left(\left[0 + h\frac{\sqrt{3}}{2}\right], \left[0 + h\left(\frac{1}{2}\right)\right]\right) - f(0,0)}{h}$$

$$\Rightarrow \frac{\left(\frac{h\sqrt{3}}{2}\right)^2 \left(\frac{h}{2}\right)}{\left(\frac{h\sqrt{3}}{2}\right)^4 + \left(h\left(\frac{1}{2}\right)\right)^2} - 0$$

$$\lim_{h \rightarrow 0}$$

$$\frac{h^3 \left(\frac{3}{8}\right)}{h^2 \left[h^2 \left(\frac{9}{4}\right) + \frac{1}{4}\right]}$$

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$$\Rightarrow \frac{h^3 \left(\frac{3}{8}\right)}{h^2 \left(h^2 \frac{9}{4} + \frac{1}{4}\right)}$$

$$\Rightarrow \frac{3/8}{1/4} = \frac{3}{2} = 1.5$$

ungraded

Q. Suppose d is the directional derivative of a scalar valued multivariable function $f(x, y) = x^2 + y^2$ at a point $P(1, 2)$ in the direction of the line PQ , where Q is at $(4, 6)$. The value of d is.

Sol.

$$\overrightarrow{PQ} = (4, 6) - (1, 2)$$

$$\text{unit vector: } \left(\frac{3}{5}, \frac{4}{5} \right)$$

$$\text{gradient: } f(x, y) = x^2 + y^2$$

$$\nabla f = (2x, 2y)$$

$$\text{at } P(1, 2): \nabla f(1, 2) = (2, 4)$$

$$D_u f = \nabla f \cdot u$$

$$= (2, 4) \cdot \left(\frac{3}{5}, \frac{4}{5} \right) = \frac{22}{5} = 4.4$$